

Bondora Loan Risk & Performance Analysis

Project Objective:

The primary objective of this project is to **predict financial risk and loan defaults** for the Bondora peer-to-peer (P2P) lending platform. By analyzing borrower demographics, creditworthiness, loan statuses, repayment behavior, overdue trends, and recovery performance, the project aims to provide a comprehensive understanding of loan performance and associated risks. This enables accurate identification of high-risk borrowers, enhances credit decision-making, and strengthens portfolio management strategies.

About Bondora & Importance of Risk Assessment

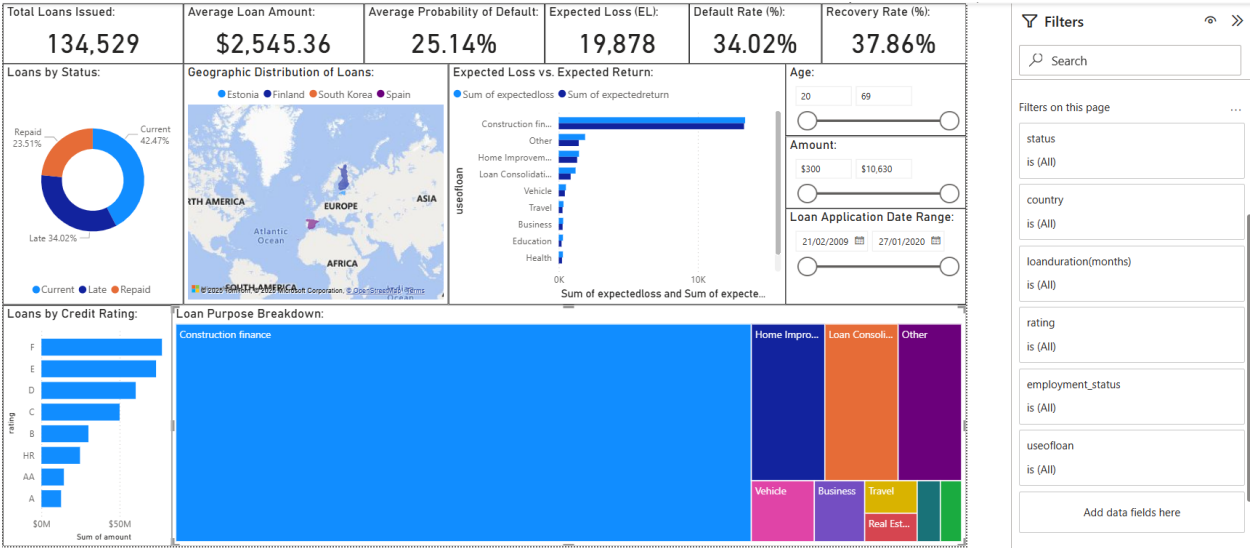
Bondora is a leading European P2P lending platform that connects investors with borrowers seeking personal loans. Investors earn returns based on interest payments, while borrowers gain access to fast and flexible financing.

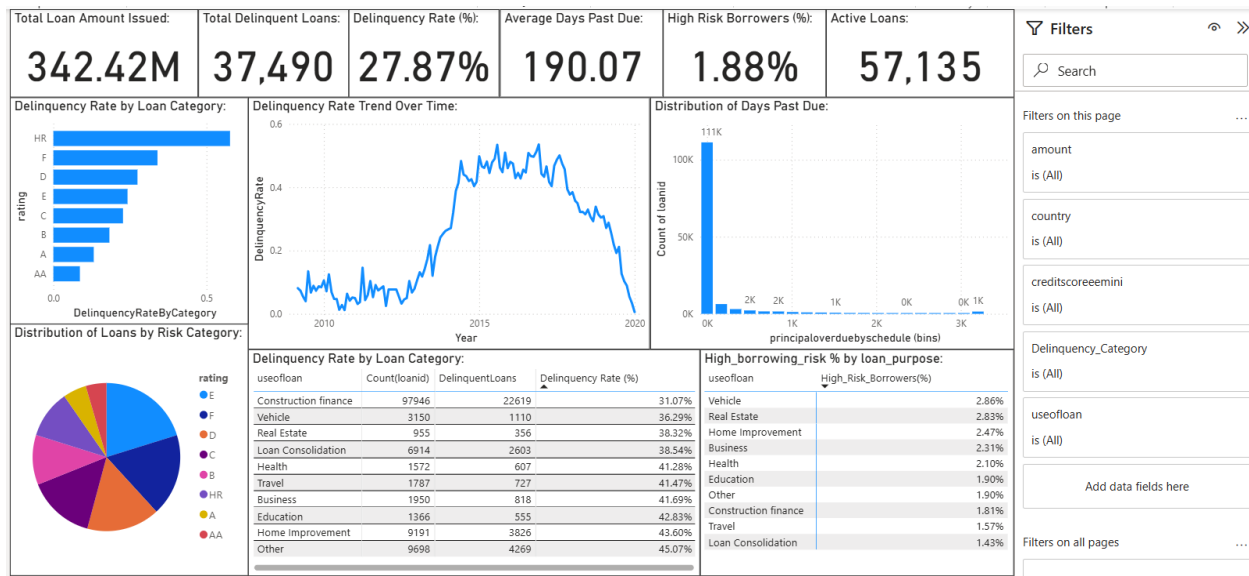
In such a lending ecosystem, **risk assessment holds a crucial role** because:

- Borrowers come from varied financial and demographic backgrounds.
- Loan defaults directly affect investor earnings.
- Understanding repayment habits, delinquency patterns, and recovery rates is essential for maintaining portfolio health.
- Proper risk segmentation allows Bondora to optimize credit policies and loan pricing.

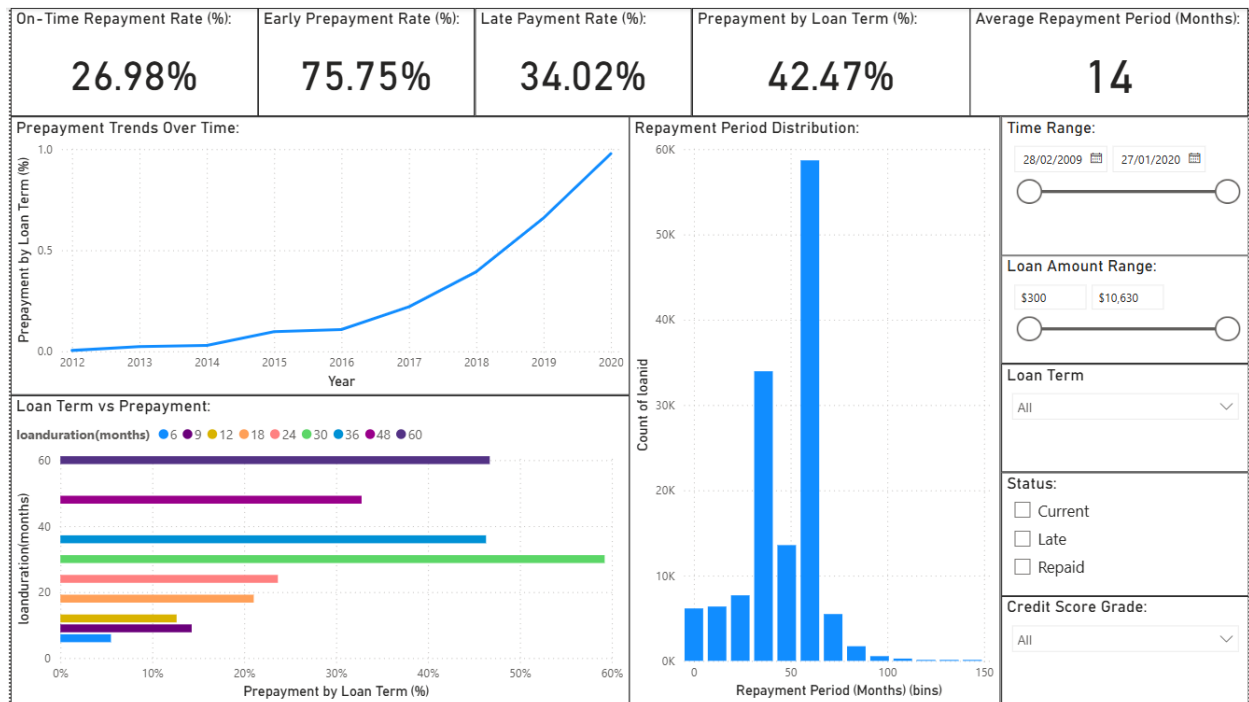
Accurate prediction of financial risk helps minimize defaults, improve collection strategies, and maintain strong investor confidence.

Dashboard Summary & Key Insights:

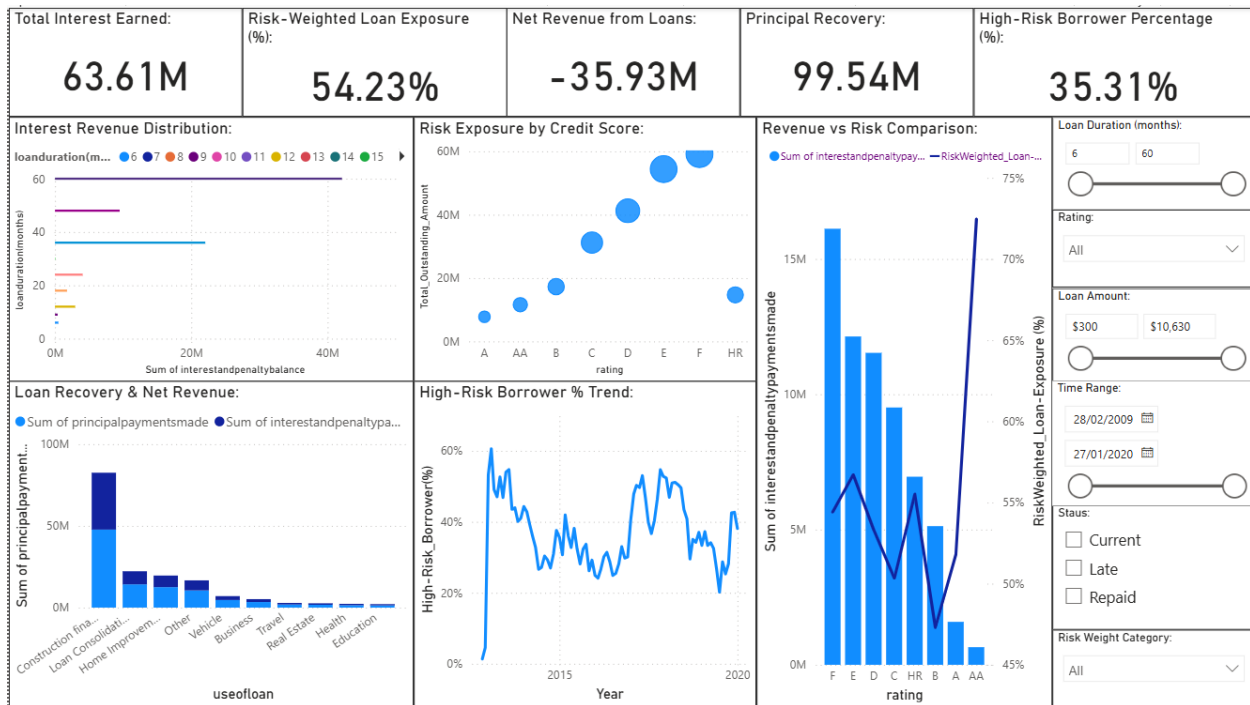




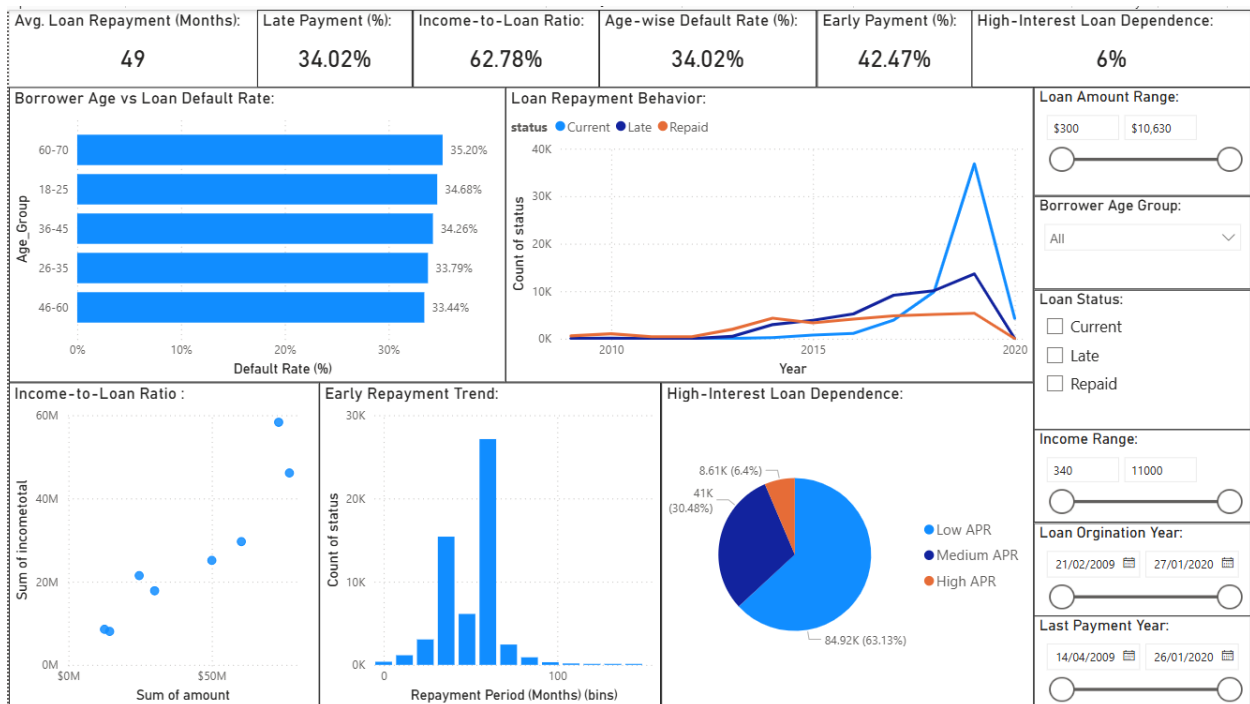
Borrower Risk & Loan Delinquency Analysis



Loan Repayment & Prepayment Analysis



Loan Profitability and Risk Analysis



Borrower Behavior & Loan Performance Analysis

1. Loan Portfolio Analysis:

Loan Volume, Composition & Distribution:

- **Total Loan Amount Issued:** ≈ **\$342M** (large and diversified portfolio).
- **Average Loan Amount:** ≈ **\$2,545**, suggesting a focus on **small to medium consumer loans**.
- **Loan Purpose Distribution:**
 - *Construction Finance, Home Improvement, Loan Consolidation, and other* categories form the bulk of loan volume.
 - Construction finance alone dominates several visuals (both loan amount & risk charts).

Loan Status Breakdown:

- **Current Loans:** ~42–43%
- **Late Loans:** ~34%
- **Repaid Loans:** ~23%
 - High late loan proportion signals substantial performance friction.

Trends in Loan Issuance Over Time:

- Consistent loan issuance from 2009–2020 with occasional spikes.
- Loan volumes increased significantly around 2015–2017 before stabilizing.
- Growth coincides with higher risk exposure years (seen in credit score & delinquency trends).

2. Borrower Demographics & Creditworthiness:

Age-Based Insights:

- Default rates are **stable across age groups** (33–35%), with the **60–70 group slightly higher**.
- Borrowers aged **26–45** form the largest share of borrowers and loan value.

Income-to-Loan Trends:

- Portfolio-wide **income-to-loan ratio ~62–63%**, a moderate buffer.
- Higher loan amounts generally correspond to higher incomes, but the relationship is non-linear.
- Low-income borrowers tend to dominate the high-risk category.

Employment & Credit Score:

- Credit rating categories E, F, HR contribute a disproportionate share of:
 - Total outstanding debt
 - Default and delinquency rates
 - Risk-weighted loan exposure
- High-risk borrowers account for **~35% of the portfolio**, a significant concentration.

High-Risk Borrower Patterns

- High-risk percentage peaked around **2013–2016**, then began declining.
- High-risk borrowers often have: Lower or unstable income, longer repayment terms, Higher APR loans

3.Loan Performance & Default Trends:

Default Rate:

- Overall default rate hovers around **34%**—substantial and alarming.
- Default is strongly correlated with:
 - **Longer loan terms**
 - **Lower credit ratings (HR, F, E)**
 - **Certain loan categories**
 - Highest default categories: *Home Improvement, Education, Business, Health, Travel.*

Delinquency Trends:

- Delinquency rates peaked around **2013–2015**, then improved.
- Many delinquent accounts fall in the **<30 days** bucket ($\approx 64\%$), suggesting partial recoverability.

Factors Leading to High Default Risk:

- Low credit score + high loan amount
- High APR loans
- Certain loan purposes (home improvement, business)
- Borrowers with income-to-loan ratios $< 60\%$
- Longer repayment terms (36–60 months)

4. Financial Risk Prediction Model:

Default Risk Prediction Accuracy:

- Not explicitly quantified in dashboards, but model-derived metrics like:
 - **Probability of Default (PD) ~25% (avg)**
 - **Expected Loss (EL) $\approx$ 19,878**

Indicate a functioning predictive risk scoring system.

Key Inputs Influencing Approval & Risk Scoring:

The dashboards highlight predictors frequently aligned with default:

- **Credit score category**
- **Loan duration**
- **Income-to-loan ratio**
- **Loan purpose**
- **Borrower age**
- **Repayment history**
- **APR category**

These likely form the core of the risk model for PD and EL.

5. Repayment Behavior & Late Payment Trends:

Repayment Patterns:

- **Average repayment period $\approx$ 49 months** (loan book skewed toward long-term loans).
- Bulk of repayments clusters around **36–60 months**, signaling borrower preference for longer terms.
- Average repayment period reported in one dashboard drops to **14 months** (likely for early re-payers only).

Late Payment Insights

- **Late Payment Rate:** $\sim$ 34%
- Late payments correlate with:
 - High APR loans
 - Low credit ratings
 - Low-income borrowers

Early Payment Trends

- Early payment rate surprisingly high at **42–75%** depending on subset.
- Strong early payment observed for:
 - Short loan terms
 - High-income borrowers
 - Better credit ratings (A–C)

6. Profitability & Risk Exposure Analysis:

Interest & Revenue:

- **Total Interest Earned:** ~63.61M
- **Net Revenue from Loans: –35.93M (loss)**
This indicates **losses driven by loan defaults outweigh interest income**, exposing a structural risk issue.

Outstanding vs. Recovered:

- **Principal Recovery Rate:** ~41–42%
- Outstanding amount remains high at **~\$237M**, limiting liquidity.

Risk Exposure Patterns:

- Risk-weighted exposure concentrated in:
 - Credit scores **E, F, HR**
 - Loan purposes: Construction finance, home improvement, business

ROI vs Default Risk:

- Lower credit categories produce higher interest revenues (expected), **but default erosion is far greater**, creating negative ROI.
- High-risk categories have poor recovery rates (**1–1.8%**), further reducing investor returns.

Final Strategic Insights:

The analysis shows that Bondora's loan portfolio is mainly struggling because of the **high number of defaults and late payments**. These problem areas are pulling down overall performance. Even though high-risk borrowers bring in more interest, the added risk doesn't pay off—the **losses from defaults outweigh the gains**, creating an unfavorable risk–return balance.

To strengthen portfolio performance, Bondora should **tighten approval standards** for E–HR borrowers and **revisit how long-term loans are priced and assessed**, since these tend to fail more often. Improving underwriting will also make a big difference—this includes using stricter income-to-loan ratios, digging deeper into loan purposes that often lead to default, and using predictive analytics to catch early warning signs before loans turn risky.

For better stability, Bondora should focus on **recovering loans that are under 30 days past due**, as this group is the largest and offers the best chance of recovering funds quickly. Encouraging customers to choose **shorter loan durations** can also reduce long-term risk exposure. In the long run, incorporating broader **economic factors**—such as employment trends and inflation—into the risk model will help Bondora build a stronger, more forward-looking lending strategy.

Key Business Implications:

1. What actions should Bondora take based on the insights?

- Reduce lending to high-risk (E–HR) borrowers to limit default exposure.
- Revisit pricing and approval rules for long-term loans, which show higher chances of default.
- Strengthen early recovery efforts, especially for loans <30 days overdue, where quick action has the highest impact.
- Encourage shorter loan durations through incentives to lower long-term repayment risk.

2. Suggestions for improving loan approval processes:

- Apply stricter income-to-loan ratio criteria to ensure borrowers can comfortably repay.
- Assess loan purpose more carefully, especially those linked to higher default rates.
- Use predictive analytics to flag early warning signs before a loan becomes risky.
- Adopt a multi-layered underwriting approach, considering credit history, financial stability, and economic context.
- Standardize documentation and verification to avoid inconsistencies and approval of errors.

3. Strategies for reducing defaults and increasing investor confidence:

- Act quickly on early delinquencies and provide repayment guidance to improve recovery and keep borrowers on track.
- Continuously monitor portfolio risk using real-time dashboards and alerts.
- Adjust risk-based pricing to match borrower profiles and diversify the portfolio to reduce concentration risk.
- Communicate transparently with investors about default trends, recovery performance, and model improvements.

Conclusion & Recommendations:

Key Takeaways from the Analysis:

- High default and delinquency rates—mainly from E–HR borrowers and long-term loans—are the key drivers of portfolio underperformance.
- High-risk loans generate interest but still create a **negative risk–return balance** due to frequent defaults.
- Early-stage delinquencies (<30 days) form the largest group and offer the most potential for quick recovery.
- Stronger underwriting and better borrower segmentation are essential for improving portfolio quality.

Final Recommendations for Bondora’s Risk Management Team:

Tightening lending rules for **high-risk borrowers** reduces **default exposure** and strengthens **portfolio health**. **Repricing** long-term loans based on actual **risk** ensures fair and accurate interest rates. Stricter **underwriting**—through tighter **income-to-loan checks** and clearer **loan-purpose evaluation**—improves credit decisions. Strengthening **early recovery actions** for **<30-day delinquencies** boosts **recoveries** and prevents further slippage, creating a more **resilient credit risk framework**.

Future Scope: Enhancing Financial Risk Prediction:

- Adopt advanced machine learning models to increase default prediction accuracy.
- Integrate macroeconomic factors (employment, inflation) and behavioral data into risk models.
- Build real-time monitoring dashboards with early-warning alerts.
- Develop stress-testing frameworks to understand how portfolio risk changes under different economic scenarios.